

Microbes In Human Welfare

INTRODUCTION

Microbes : Major Component of biological systems



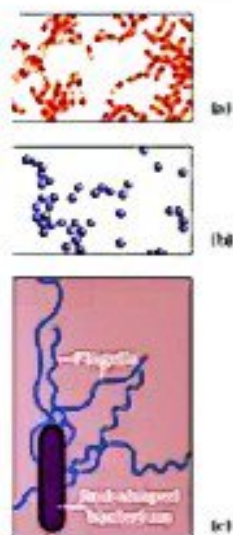
Are Diverse

- (1) microscopic animals & Plant viruses
- (2) Viroid's.
- (3) Prions (proteinaceous Infectious agent)
- (4) Fungi
- (5) Protozoa

Microbes Present Everywhere :

- Soil
- Water
- Air
- Inside our bodies
- Other plants and animals
- Deep inside geysers
- Under layers of snow
- Highly acidic environments





10.1 Bacteria: (a) Rod-shaped, $\times 1500$; (b) Spherical, magnified $\times 15000$; (c) A rod-shaped bacterium showing flagella, $\times 150,000$

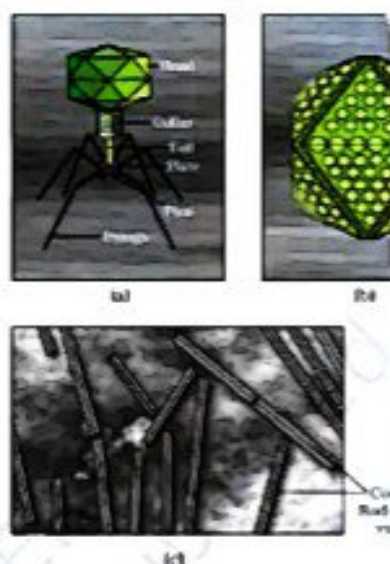
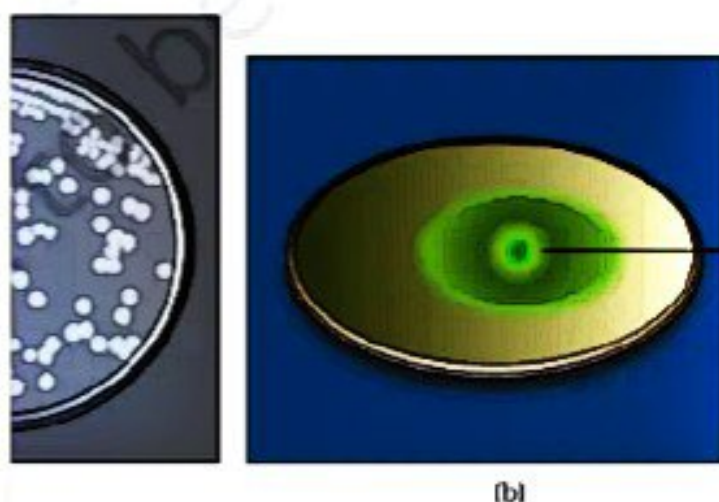


Figure 10.2 Viruses: (a) A bacteriophage Adenovirus which causes respiratory infections; (b) Rod-shaped Mosaic Virus (TMV). Magnified $1,00,000-1,50,000\times$



10.3 (a) Colonies of bacteria growing in a petri dish; (b) Fungal colony growing in a petri dish

Figure-10.1, Figure-10.2, Figure-10.3

Bacteria + Many fungi $\rightarrow$ grown on Nutritive media



Form Colonies

MICROBES IN HOUSEHOLD PRODUCTS

Lactic acid bacteria Eg : Lactobacillus

- Converts milk $\rightarrow$ Curd
- Produces acid $\rightarrow$ Coagulates & partially digests milk



- Improves Nutritional quantity of Vitamin B12
- In stomach , checks disease causing microbes

DOUGH

- ~Used to make Idli & Dosa
- ~Fermented by Bacteria
- ~Puffed up - Due to production of CO₂

Dough → Used for making Bread



Fermented By yeast - Saccharomyces Cerevisiae



Baker's Yeast

TODDY

- ~Traditional drink - In southern Counties
- ~Made by fermenting sap from palms
- ~Ferment fish , Soyabean & bamboo shoots

To make foods



CHEESE

~Oldest food items

~Varieties - on the basis of characteristic texture,
Flavour, Taste

Examples Of Cheese :

Swiss Cheese - *Propionibacterium Sharmanii*
(Bacteria)

Roquefort Cheese - *Penicillium Roqueforti* (fungi)

|

Imparts particular flavour

MICROBES IN INDUSTRIAL PRODUCTS

FERMENTOUS → Large Vessels

↓

Used for production on an industrial scale





Figure 10.4 Fermentors



Figure 10.5 Fermentation Plant

Figure-10.4

Figure-10.5

FERMENTED BEVERAGES

YEAST - (*Saccharomyces cerevisiae*) Used for Bread making & Commonly called **Brewer's yeast**

~Produce : Ethanol

~Uses : Fermenting Malted cereals & fruit juices

Types Of Alcoholic Drinks

(Type of raw material used & type of processing)



Without Distillation

- Beer
- Wine

Distillation

- Whisky
- Brandy
- Rum

ANTIBIOTICS

~Anti + Bio - Against + life (Wrt microbes) ↴
But pro life with respect to Humans

- ~Produced my Microbes
- ~Significant Discovery of 20th Century

Chemical Substances : Kills / retards growth of disease causing microbes

↴ Change Discovery (By Alexander Fleming)

Eg : Penicillin

↴ First antibiotic

Alexander - Working on Staphylococci Bacteria



Observed Mould growing
(Penicillium notatum)



Due to chemical ← Around this Staphylococci
↓ Couldn't Grow

Named it Penicillin

Fully Potential – By Erbest Chain & Howard Florey
–Used extensively to treat American soldiers



During World war II

–Fleming, Chain, Florey – Noble prize in 1945
–Have improved our capacity to treat deadly
disease :

- (a) Leprosy (Kusht rog)
- (b) Whooping Cough (Kali khansi)
- (c) Diphtheria (Gal ghotu)
- (d) Plague



Chemicals , Enzymes & Other Bioactive Molecules

ACID PRODUCERS ARE :

- Aspergillus Nigar (a fungus) - Citric acid
- Acetobactor aceti (a bacterium) - Acetic acid
- Clostridium Butylicum (bacterium) - Butyric acid
- Lactobacillus (bacterium) - Lactic acid

LIPASES

- Used in detergent formulations
- Helpful in removing Oily Stains

PECTINSASES AND PROTEASES

- Used to Clarify Bottled Juices



STREPTOKINASE

- From Streptococcus (bacterium)
- Used as 'Clot Buster' → for patients undergoing myocardial infraction

CYCLOSPORIN A

- Used as immunosuppressive agents



In organ Transplantation

- Produced by Trichoderma polysporum – Fungus

STATINS

- As Blood Cholesterol lowering agent
- Competitively inhibits enzyme responsible for Cholesterol synthesis
- By yeast : Monascus purpureus

MICROBES IN SEWAGE TREATMENT

Municipal waste water → Sewage



Major Component → Human Excreta

Contains large amount of organic matter & microbes

STPs (Sewage treatment plants) – make sewage
| less polluting

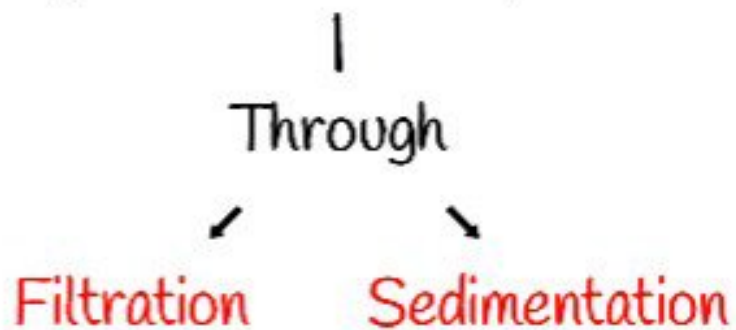
By heterotrophic microbes present naturally

Treatment – two Stages

(a) Primary treatment

(b) Secondary treatment

Primary Treatment – Physical removal of particles



Floating debris removed by sequential filtration

Grit removed By sedimentation

Solids settles (Primary sludge)



↓
Supernatant (effluent) form

↓
Effluent taken for 2 degree treatment

Secondary / Biological Treatment :

Primary effluent – Large aeration tanks – **Agitated mechanically**

↓
Air pumped into it

↓
Allows vigorous growth of microbes
(Aerobic)

✓
Consumes major
part of organic matter

↘
into **flocs**
(bacteria associated with
fungal filaments to form
Mesh)

↓
Passed into settling tank

Bacterial flocs settle down
(Activated sludge)



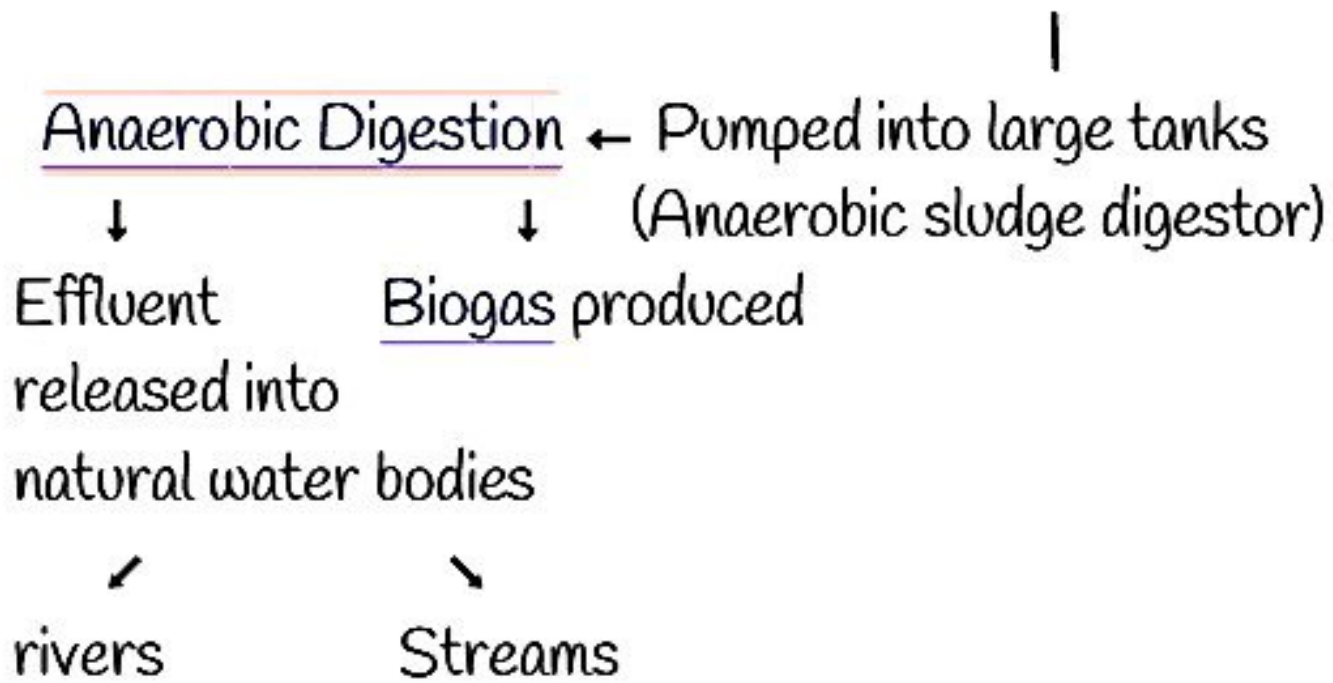


Figure 10.6 Secondary treatment

Figure-10.6



BIOCHEMICAL OXYGEN DEMAND (BOD)

~Amount of O_2 that would be consumed if all the organic matter in one litre of H_2O were oxidized by bacteria

- Measures rate of uptake of O_2 by microorganisms
- Measure of organic matter present in water
- Greater the BOD, more is its polluting potential

~Effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment & the sediment is called **Activated Sludge**

~Small part of the activated sludge is pumped back into the aeration tank to serve as the inoculum

~Remaining major part of the sludge is pumped into large tanks called **Anaerobic Sludge digesters**

Other kinds of bacteria grow anaerobically



|
Digest the bacteria & the fungi in Sludge

BIOGAS - Mixture of CH_4 , H_2 & CO_2

CH_4 : (50-70%)

H_2 : (in traces)

CO_2 : (30-40%)

Ganga action plan (GAP) Yamuna action plan (YAP)

|
Initiated by ministry of Environment forests

PROPOSAL :

To build a large no. Of sewage treatment plants such that only treated sewage may be discharged into rivers

MICROBES IN PRODUCTION OF BIOGAS

~Biogas is a mixture of gases (containing predominantly



methane) produced by the microbial activity

~Microbes – produce gas – Depending on the organic substrate they use

METHANOGENS

- Grows anaerobically on cellulose material
- Commonly found in anaerobic sludge (Sewage treatment)
- Produce $\text{CH}_4 + \text{CO}_2 + \text{H}_2$
- Present in rumen of cattle (provides nutrition by breaking down of cellulosic material)

Eg : Methanobacterium

Found in dung – Used for the production of Biogas

BIOGAS PLANT

Concrete Tank (10-15 deep)



Biowaste Collected – Slurry cover is placed



Keeps on rising as gas is produced in tank



Spent slurry removed



Used as fertiliser

Biogas plant – Has an outlet



Connected to pipe to supply biogas to houses

Technology Developed in Indian Due to



IARI

Indian agricultural
research institute



KVIC

Khadi & Village
industries commission

~Biogas Plant – Mainly found in rural areas

~Biogas Used for :

- Lightening
- Cooking



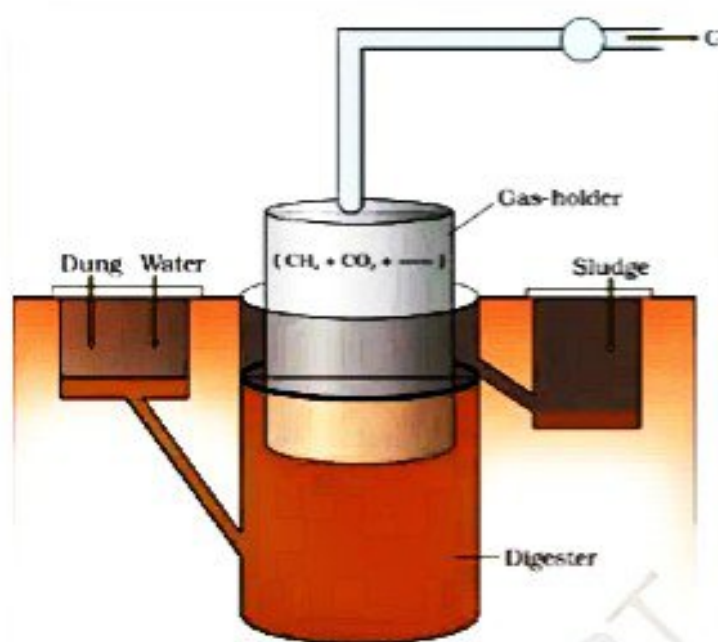


Figure 10.8 A typical biogas plant

Figure-10.8

MICROBES AS BIOCONTROL AGENTS

Biocontrol – Use of biocontrol methods to control plant diseases and pests

In modern society, we use Insecticides, pesticides & weedicides

↓
↻ Chemicals ↻

Pollute environment
(Soils, Ground water)

Toxic, extremely
harmful





Soil polluted through over use of Weedicides

CONTROL OF PESTS & DISEASES

In agriculture, Pest control - Relies on natural predators rather than introduced chemicals

ORGANIC FARMING

~Belief - Biodiversity furthers health

~More variety, more sustainable it is

~Pests are not eradicated, kept at a manageable levels by
- System of checks & balance

~Eradication of pests - Undisirable because depending predator & parasitic bird won't survive

Conventional Farming Practices

Use chemical methods to kill useful & harmful life forms indiscriminately

Biological farming practices

✓ Various life forms inhabit field

Helps in Knowing



↘ Predators & pests

- Life cycle
- Patterns of breeding
- Habitat

↗ Beetle with red & black markings

Ladybird

↘ Used to get rids of aphids

Dragonfly - to get rid of mosquito

Bacillus Thuringiensis (BT)

~Insect resistant plant

|

By BT cotton (Unique technique of genetic engineering)

- Introduced to control butterflies & Caterpillars
- Available as dry spores (mixed C & H₂O & sprayers)
- Sprayed on brassicas & fruit trees

|



eating by insects - gut release toxin - insect killed

TRICHODERMA

- Fungus
- Free living
- Common in root ecosystems
- Effective biocontrol agent

BACULOVIRUSES

- Pathogens
- Attack insects + arthropods
- Species Specific
- No negative Impact on

|

Plants, animals, birds, mammals, fishes & non-targeted insects

- Helps conserve beneficial pests

|

Aids in integrated pest management programme
(IPM)

- Narrow spectrum
- Belongs to Genus **Nucleopolyhedrovirus**



MICROBES AS BIOFERTILISERS

✓ Organisms – enrich nutrient quality of soil

Biofertilisers

✓ Sources

(a) Bacteria

(b) Fungi

(c) Cyanobacteria

Symbiotic Association – b/w nodules of root of leguminous plants

+

Rhizobium

~Fix atmospheric N_2 → into organic forms

~Other free living bacteria → Eg : Azospirillum

Azotobacter

Enriches nitrogen Content of Soil

MYCORRHIZA

Symbiotic association between roots of higher Plants & fungi



Synthesize food for themselves & fungi

Eg : Glomus
Absorbs phosphorous & passes to plants

BIOFERTILISERS

- Low cost input
- Don't pollute the environment
- Reduce dependence on chemical
- Used to replenish soil nutrients

CYANOBACTERIA

- Autotrophic microbes
- Widely distributed in aquatic & terrestrial environments
- Many fix N_2
- Eg : Anabaena, Nostoc, Oscillatoria

~Important Biofertiliser - in paddy field

~Blue green algae - add organic material to soil

Increases it's fertility

